28. Suppose n denotes a nonnegative integer. Determine the values of n such that $z^n = 1$ possesses only real solutions. Defend your answer with sound mathematics.

29. (**a**) Proceed as in Example 2 to find the approximate values of the two square roots w_0 and w_1 of $1 + i$.

(**b**) Show that the exact values of the roots in part (a) are

$$w_0 = \sqrt{\frac{1+\sqrt{2}}{2}} + i\sqrt{\frac{\sqrt{2}-1}{2}}, \quad w_1 = -\sqrt{\frac{1+\sqrt{2}}{2}} - i\sqrt{\frac{\sqrt{2}-1}{2}}.$$

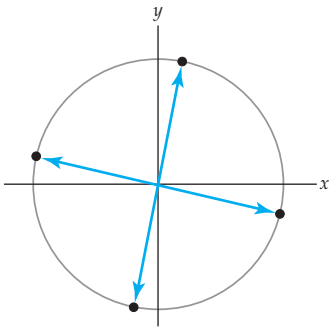


FIGURE 1.4.4 Figure for Problem 34

30. Discuss: What geometric significance does the result in Problem 24 have?

31. Discuss: A real number can have a complex nth root. Can a nonreal complex number have a real nth root?

32. Suppose w is located in the first quadrant and is a cube root of a complex number z. Can there exist a second cube root of z located in the first quadrant? Defend your answer with sound mathematics.

33. Suppose z is a complex number that possesses a fourth root w that is neither real nor pure imaginary. Explain why the remaining fourth roots are neither real nor pure imaginary.

34. Discuss: The vectors in FIGURE 1.4.4 represent the four fourth roots of a complex number z. Can you determine from this figure alone which root is the principal fourth root of z?

Computer Lab Assignments

In Problems 35–40, use a CAS to first find $z^n = w$ for the given complex number and the indicated value of n. Then, using the output and the same value of n, determine whether $w^{1/n} = (z^n)^{1/n} = z$. If not, explain why not.

35. $z = 2.5 - i;\ n = 10$

36. $z = -0.5 + 0.3i;\ n = 5$

37. $z = 1 + 3i;\ n = 8$

38. $z = 2 + 2i;\ n = 12$

39. $z = i;\ n = 21$

40. $z = -1 + \sqrt{3}\,i;\ n = 11$

1.5 Sets of Points in the Complex Plane

INTRODUCTION In the preceding sections we examined some rudiments of the algebra and geometry of complex numbers. But we have barely scratched the surface of the subject known as complex analysis; the main thrust of our study lies ahead. Our goal in the chapters that follow is to examine functions of a single complex variable $z = x + iy$ and the calculus of these functions.

Before introducing the notion of a function in Chapter 2, we need to state some essential definitions and terminology about sets in the complex plane.

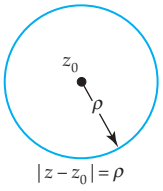


FIGURE 1.5.1 Circle of radius ρ

Circles Suppose $z_0 = x_0 + iy_0$. Since $|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$ is the distance between the points $z = x + iy$ and $z_0 = x_0 + iy_0$, the points $z = x + iy$ that satisfy the equation

$$|z - z_0| = \rho, \quad \rho > 0, \tag{1}$$

lie on a **circle** of radius ρ centered at the point z_0. See FIGURE 1.5.1.

EXAMPLE 1 Two Circles

(a) $|z| = 1$ is an equation of a unit circle centered at the origin.

(b) By rewriting $|z - 1 + 3i| = 5$ as $|z - (1 - 3i)| = 5$, we see from (1) that the equation describes a circle of radius 5 centered at the point $z_0 = 1 - 3i$. ☰

Disks and Neighborhoods The points z that satisfy the inequality $|z - z_0| \leq \rho$ can be either *on* the circle $|z - z_0| = \rho$ or *within* the circle. We say that the set of points defined by $|z - z_0| \leq \rho$ is a **disk** of radius ρ centered at z_0. But the points z that satisfy the strict inequality $|z - z_0| < \rho$ lie within, and not on, a circle of radius ρ centered at the point z_0. This set is called a **neighborhood** of z_0. Occasionally, we will need to use a neighborhood of z_0 that also *excludes* z_0. Such a neighborhood is defined by the simultaneous inequality $0 < |z - z_0| < \rho$ and is called a **deleted neighborhood** of z_0. For example, $|z| < 1$ defines a neighborhood of the origin, whereas $0 < |z| < 1$ defines a deleted neighborhood of the origin; $|z - 3 + 4i| < 0.01$ defines a neighborhood of $3 - 4i$, whereas the inequality $0 < |z - 3 + 4i| < 0.01$ defines a deleted neighborhood of $3 - 4i$.

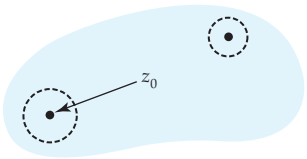


FIGURE 1.5.2 Open set

Open Sets A point z_0 is said to be an **interior point** of a set S of the complex plane if there exists some neighborhood of z_0 that lies entirely within S. If every point z of a set S is an interior point, then S is said to be an **open set**. See FIGURE 1.5.2. For example, the inequality $\text{Re}(z) > 1$ defines a *right half-plane*, which is an open set. All complex numbers $z = x + iy$ for which $x > 1$ are in this set. If we choose, for example, $z_0 = 1.1 + 2i$, then a neighborhood of z_0 lying entirely in the set is defined by $|z - (1.1 + 2i)| < 0.05$. See FIGURE 1.5.3. On the other hand, the set S of points in the complex plane defined by $\text{Re}(z) \geq 1$ is *not* open because every neighborhood of a point lying on the line $x = 1$ must contain points in S and points not in S. See FIGURE 1.5.4.

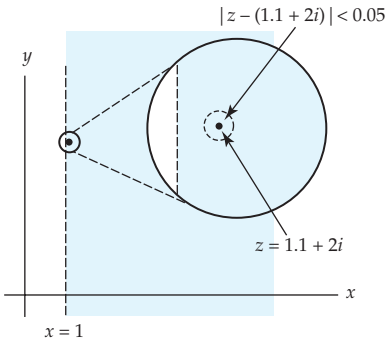


FIGURE 1.5.3 Open set with magnified view of point near $x = 1$

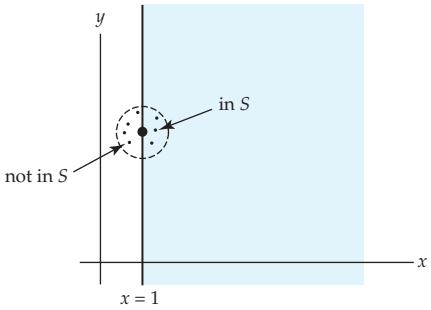
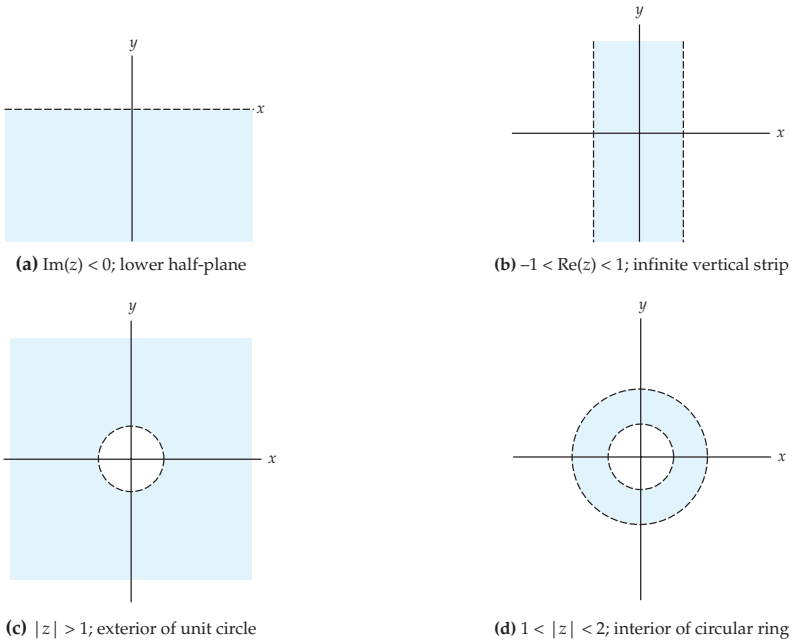


FIGURE 1.5.4 Set S not open

EXAMPLE 2 Some Open Sets

FIGURE 1.5.5 illustrates some additional open sets. ☰

If *every* neighborhood of a point z_0 of a set S contains at least one point of S and at least one point not in S, then z_0 is said to be a **boundary point** of S. For the set of points defined by $\text{Re}(z) \geq 1$, the points on the vertical line $x = 1$ are boundary



(a) Im(z) < 0; lower half-plane

(b) –1 < Re(z) < 1; infinite vertical strip

(c) |z| > 1; exterior of unit circle

(d) 1 < |z| < 2; interior of circular ring

FIGURE 1.5.5 Four examples of open sets

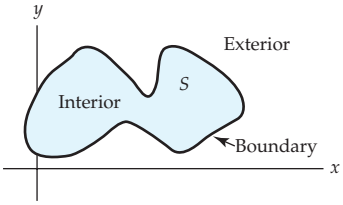


FIGURE 1.5.6 Interior, boundary, and exterior of set S

points. The points that lie on the circle $|z - i| = 2$ are boundary points for the disk $|z - i| \leq 2$ as well as for the neighborhood $|z - i| < 2$ of $z = i$. The collection of boundary points of a set S is called the **boundary** of S. The circle $|z - i| = 2$ is the boundary for both the disk $|z - i| \leq 2$ and the neighborhood $|z - i| < 2$ of $z = i$. A point z that is neither an interior point nor a boundary point of a set S is said to be an **exterior point** of S; in other words, z_0 is an exterior point of a set S if there exists *some* neighborhood of z_0 that contains no points of S. FIGURE 1.5.6 shows a typical set S with interior, boundary, and exterior.

An open set S can be as simple as the complex plane with a single point z_0 deleted. The boundary of this "punctured plane" is z_0, and the only candidate for an exterior point is z_0. However, S has no exterior points since no neighborhood of z_0 can be free of points of the plane.

Annulus The set S_1 of points satisfying the inequality $\rho_1 < |z - z_0|$ lie exterior to the circle of radius ρ_1 centered at z_0, whereas the set S_2 of points satisfying $|z - z_0| < \rho_2$ lie interior to the circle of radius ρ_2 centered at z_0. Thus, if $0 < \rho_1 < \rho_2$, then the set of points satisfying the simultaneous inequality

$$\rho_1 < |z - z_0| < \rho_2, \tag{2}$$

is the intersection of the sets S_1 and S_2. This intersection is an open circular ring centered at z_0. Figure 1.5.5(d) illustrates such a ring centered at the origin. The set defined by (2) is called an open **circular annulus**. By allowing $\rho_1 = 0$, we obtain a deleted neighborhood of z_0.

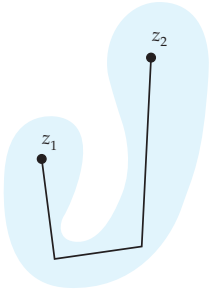


FIGURE 1.5.7 Connected set

Domain If any pair of points z_1 and z_2 in a set S can be connected by a polygonal line that consists of a finite number of line segments joined end to end that lies entirely in the set, then the set S is said to be **connected**. See FIGURE 1.5.7. An open connected set is called a **domain**. Each of the open sets in Figure 1.5.5 is connected and so each is a domain. The set of numbers z satisfying $\text{Re}(z) \neq 4$ is an open set but is not connected since it is not possible to join points on either side of

the vertical line $x = 4$ by a polygonal line without leaving the set (bear in mind that the points on the line $x = 4$ are not in the set). A neighborhood of a point z_0 is a connected set.

Regions A **region** is a domain in the complex plane together with all, some, or none of its boundary points. Since an open connected set does not contain any boundary points, it is automatically a region. A set that contains all of its boundary points is said to be **closed**. The disk defined by $|z - z_0| \leq \rho$ is an example of a closed region and is referred to as a **closed disk**. A neighborhood of a point z_0 defined by $|z - z_0| < \rho$ is an open set or an open region and is said to be an **open disk**. If the center z_0 is deleted from either a closed disk or an open disk, the regions defined by $0 < |z - z_0| \leq \rho$ or $0 < |z - z_0| < \rho$ are called **punctured disks**. *A punctured open disk is the same as a deleted neighborhood of* z_0. A region can be neither open nor closed; the annular region defined by the inequality $1 \leq |z - 5| < 3$ contains only some of its boundary points (the points lying on the circle $|z - 5| = 1$), and so it is *neither open nor closed*. In (2) we defined a circular annular region; in a more general interpretation, an annulus or **annular region** may have the appearance shown in FIGURE 1.5.8.

Note: "Closed" does not mean "not open." ▶

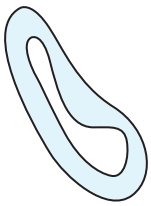


FIGURE 1.5.8 Annular region

Bounded Sets Finally, we say that a set S in the complex plane is **bounded** if there exists a real number $R > 0$ such that $|z| < R$ for every z in S. That is, S is bounded if it can be completely enclosed within *some* neighborhood of the origin. In FIGURE 1.5.9, the set S shown in color is bounded because it is contained entirely within the dashed circular neighborhood of the origin. A set is **unbounded** if it is not bounded. For example, the set in Figure 1.5.5(d) is bounded, whereas the sets in Figure 1.5.5(a), 1.5.5(b), and 1.5.5(c) are unbounded.

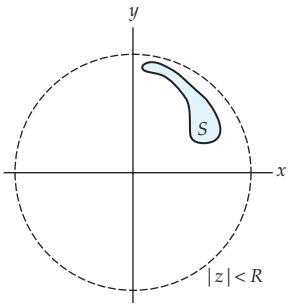


FIGURE 1.5.9 The set S is bounded since some neighborhood of the origin encloses S entirely

Remarks COMPARISON WITH REAL ANALYSIS

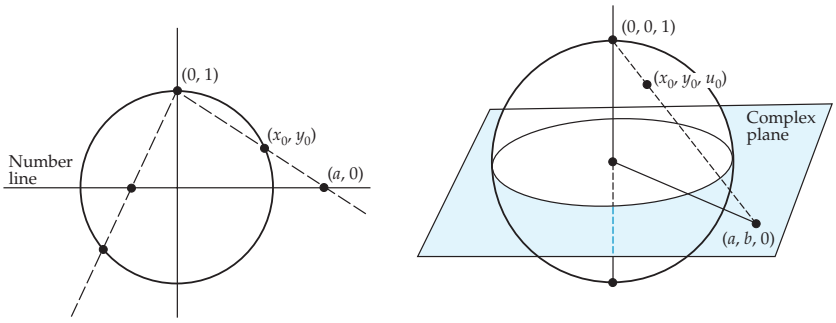
In your study of mathematics you undoubtedly found that you had to deal with the concept of *infinity*. For example, in a course in calculus you dealt with *limits at infinity*, where the behavior of functions was examined as x either increased or decreased without bound. Since we have exactly two directions on a number line, it is convenient to represent the notions of "increasing without bound" and "decreasing without bound" symbolically by $x \to +\infty$ and $x \to -\infty$, respectively. It turns out that we can get along just fine without the designation $\pm\infty$ by dealing with an "ideal point" called the **point at infinity,** which is denoted simply by ∞. To do this we identify any real number a with a point (x_0, y_0) on a unit circle $x^2 + y^2 = 1$ by drawing a straight line from the point $(a, 0)$ on the x-axis or horizontal number line to the point $(0, 1)$ on the circle. The point (x_0, y_0) is the point of intersection of the line with the circle. See Problem 47 in Exercises 1.5. It should be clear from FIGURE 1.5.10(a) that the farther the point $(a, 0)$ is from the origin, the nearer (x_0, y_0) is to $(0, 1)$. The only point on the circle that does not correspond to a real number a is $(0, 1)$. To complete the correspondence with *all* the points on the circle, we identify $(0, 1)$ with ∞. The set consisting of the real numbers $\mathbf{R}$ adjoined with ∞ is called the **extended real-number system**.

In our current study the analogue of the number line is the complex plane. Recall that since $\mathbf{C}$ is not ordered, the notions of z either "increasing" or "decreasing" have no meaning. However, we know that by increasing the modulus $|z|$ of a complex number z, the number moves farther from the origin. If we

allow $|z|$ to become unbounded, say, along the real and imaginary axes, we do not have to distinguish between "directions" on these axes by notations such as $z \to +\infty$, $z \to -\infty$, $z \to +i\infty$, and $z \to -i\infty$. In complex analysis, only a single notion of ∞ is used because we can extend the complex number system **C** in a manner analogous to that just described for the real number system **R**. This time, however, we associate a complex number with a point on a unit sphere called the **Riemann sphere**. By drawing a line from the number $z = a + ib$, written as $(a,\ b,\ 0)$, in the complex plane to the north pole $(0,\ 0,\ 1)$ of the sphere $x^2 + y^2 + u^2 = 1$, we determine a unique point $(x_0,\ y_0,\ u_0)$ on a unit sphere. As can be visualized from Figure 1.5.10(b), a complex number with a very large modulus is far from the origin (0, 0, 0) and, correspondingly, the point $(x_0,\ y_0,\ u_0)$ is close to $(0,\ 0,\ 1)$. In this manner each complex number is identified with a single point on the sphere. See Problems 48–50 in Exercises 1.5. Because the point $(0,\ 0,\ 1)$ corresponds to no number z in the plane, we correspond it with ∞. Of course, the system consisting of **C** adjoined with the "ideal point" ∞ is called the **extended complex-number system**.

This way of corresponding or mapping the complex numbers onto a sphere—north pole $(0,\ 0,\ 1)$ excluded—is called a **stereographic projection**.

For a finite number z, we have $z + \infty = \infty + z = \infty$, and for $z \neq 0$, $z \cdot \infty = \infty \cdot z = \infty$. Moreover, for $z \neq 0$ we write $z/0 = \infty$ and for $z \neq \infty$, $z/\infty = 0$. Expressions such as $\infty - \infty$, ∞/∞, ∞^0, and 1^∞ cannot be given a meaningful definition and are called indeterminate.



(a) Unit circle **(b)** Unit sphere

FIGURE 1.5.10 The method of correspondence in (b) is a stereographic projection

EXERCISES 1.5

Answers to selected odd-numbered problems begin on page ANS-3.

In Problems 1–12, sketch the graph of the given equation in the complex plane.

1. $|z - 4 + 3i| = 5$

2. $|z + 2 + 2i| = 2$

3. $|z + 3i| = 2$

4. $|2z - 1| = 4$

5. $\text{Re}(z) = 5$

6. $\text{Im}(z) = -2$

7. $\text{Im}(\bar{z} + 3i) = 6$

8. $\text{Im}(z - i) = \text{Re}(z + 4 - 3i)$

9. $|\text{Re}\,(1 + i\bar{z})| = 3$

10. $z^2 + \bar{z}^2 = 2$

11. $\text{Re}(z^2) = 1$

12. $\arg(z) = \pi/4$

In Problems 13–24, sketch the set S of points in the complex plane satisfying the given inequality. Determine whether the set is (**a**) open, (**b**) closed, (**c**) a domain, (**d**) bounded, or (**e**) connected.

13. $\text{Re}(z) < -1$

14. $|\text{Re}\,(z)| > 2$

15. $\text{Im}(z) > 3$

16. $\text{Re}\,((2+i)z+1) > 0$

17. $2 < \text{Re}(z-1) < 4$

18. $-1 \le \text{Im}(z) < 4$

19. $\text{Re}(z^2) > 0$

20. $\text{Im}(z) < \text{Re}(z)$

21. $|z-i| > 1$

22. $2 < |z-i| < 3$

23. $1 \le |z-1-i| < 2$

24. $2 \le |z-3+4i| \le 5$

25. Give the boundary points of the sets in Problems 13–24.

26. Consider the set S consisting of the complex plane with the circle $|z| = 5$ deleted. Give the boundary points of S. Is S connected?

In Problems 27 and 28, sketch the set of points in the complex plane satisfying the given inequality.

27. $0 \le \arg(z) \le \pi/6$

28. $-\pi < \arg(z) < \pi/2$

In Problems 29 and 30, describe the shaded set in the given figure using $\arg(z)$ and an inequality.

29.

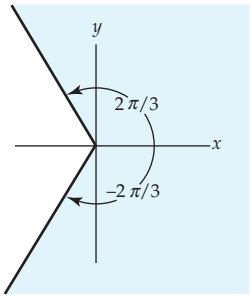


FIGURE 1.5.11 Figure for Problem 29

30.

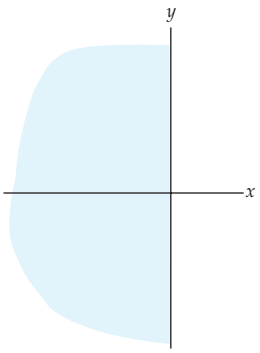


FIGURE 1.5.12 Figure for Problem 30

In Problems 31 and 32, solve the given pair of simultaneous equations.

31. $|z| = 2, \quad |z-2| = 2$

32. $|z-i| = 5, \quad \arg(z) = \pi/4$

Focus on Concepts

33. On page 29 we stated that if $\rho_1 > 0$, then the set of points satisfying $\rho_1 < |z - z_0|$ is the exterior to the circle of radius ρ_1 centered at z_0. In general, describe the set if $\rho_1 = 0$. In particular, describe the set defined by $|z + 2 - 5i| > 0$.

34. (**a**) What are the boundary points of a deleted neighborhood of z_0?

(**b**) What are the boundary points of the complex plane?

(**c**) Give several examples, not including the one given on page 30, of a set S in the complex plane that is neither open nor closed.

35. Use complex notation and inequalities in parts (a) and (b).

(**a**) Make up a list of five sets in the complex plane that are connected.

(**b**) Make up a list of five sets in the complex plane that are not connected.

36. Consider the disk centered at z_0 defined by $|z - z_0| \leq \rho$. Demonstrate that this set is bounded by finding an $R > 0$ so that all points z in the disk satisfy $|z| < R$. [*Hint*: See the discussion on inequalities in Section 1.2.]

37. Suppose z_0 and z_1 are distinct points. Using only the concept of distance, describe in words the set of points z in the complex plane that satisfy $|z - z_0| = |z - z_1|$.

38. Using only the concept of distance, describe in words the set of points z in the complex plane that satisfies $|z - i| + |z + i| = 1$.

In Problems 39 and 40, describe the shaded set in the given figure by filling in the two blanks in the set notation

$$\{\, z\text{: } _____ \text{ and/or } _____ \}$$

using complex notation for equations or inequalities and one of the words *and* or *or*.

39.

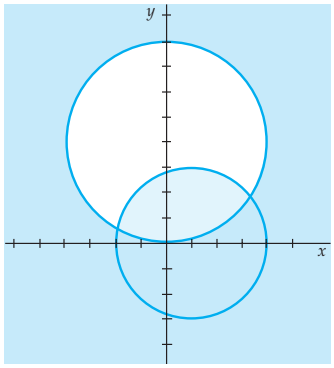


FIGURE 1.5.13 Figure for Problem 39

40.

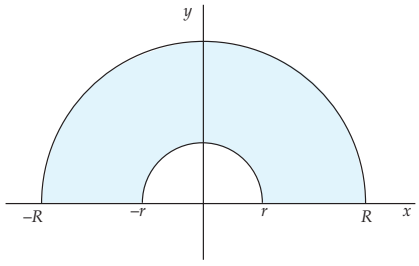


FIGURE 1.5.14 Figure for Problem 40

41. Consider the set S of points in the complex plane defined by $\{i/n\}$, $n = 1, 2, 3, \ldots$. Discuss which of the following terms apply to S: *boundary*, *open*, *closed*, *connected*, *bounded*.

42. Consider a finite set S of complex numbers $\{z_1, z_2, z_3, \ldots, z_n\}$. Discuss whether S is necessarily bounded. Defend your answer with sound mathematics.

43. A set S is said to be **convex** if each pair of points P and Q in S can be joined by a line segment $\overline{PQ}$ such that every point on the line segment also lies in S. Determine which of the sets S in the complex plane defined by the following conditions are convex.

(a) $|z - 2 + i| < 3$

(b) $1 < |z| < 2$

(c) $x > 2,\ y \leq -1$

(d) $y < x^2$

(e) $\text{Re}(z) \leq 5$

(f) $\text{Re}(z) \neq 0$

44. Discuss: Is a convex set, defined in Problem 43, necessarily connected?

45. Discuss and decide whether the empty set $\varnothing$ is an open set.

46. Suppose S_1 and S_2 are open sets in the complex plane.

(a) Discuss: Is the union $S_1 \cup S_2$ an open set? If you think the statement is true, try to prove it. If you think the statement is false, find a counterexample.

(b) Repeat part (a) for the intersection $S_1 \cap S_2$.

Before answering Problems 47–50, reread the *Remarks* on the end of this section.

47. Find the point $(x_0,\ y_0)$ on the unit circle that corresponds to each of the real numbers: $-\frac{1}{4},\ \frac{1}{2},\ -3,\ 1,\ 10$. See Figure 1.5.10(a).